

Lab 5

- (1) Show that HALT_{TM} is undecidable. (Assume we know A_{TM} is undecidable, you are required to reduce A_{TM} to HALT_{TM} .)
- (2) Show that a language is decidable iff it is Turing-recognizable and co-Turing-recognizable.
- (3) Give the definition of the following concept: Language A is mapping reducible to language B.
- (4) Show that $A_{\text{TM}} \leq_m \text{HALT}_{\text{TM}}$.
- (5) Definition: A linear bounded automaton (LBA) is a restricted type of Turing machine wherein the tape head isn't permitted to move off the portion of the tape containing the input. If the machine tries to move its head off either end of the input, the head stays where it is, in the same way that the head will not move off the left-hand end of an ordinary Turing machine's tape.
 - (a) Proof that: Let M be an LBA with q states and g symbols in the tape alphabet. There are exactly qng^n distinct configurations of M for a tape of length n .
 - (b) Let $A_{\text{LBA}} = \{ \langle M, w \rangle \mid M \text{ is an LBA that accepts input } w \}$. Proof that A_{LBA} is decidable.